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Third Semester B.E. Degree Examination, June/July 2023
Data Structures and Applications

Max. Marks: 100

Time: 3 hrs.

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define data structures. Classify data structures in various features. (06 Marks)
- b. Write algorithms to insert a data element into array and delete an element from the array. (07 Marks)
- c. Explain various memory allocation and de-allocation function supported in C. (07 Marks)

OR

- 2 a. Explain user defined structures with respect to C. Give structure definition and declaration for STUDENT data with the following information: USN and Name. Also give self referential structure. (04 Marks)
- b. Show array representation of two polynomials. Write a C function to add two polynomials A(x) and B(x) term by term to produce D(x) where $D(x) = A(x) + B(x)$, $A(x) = 2x^{10} + x + 3$, $B(x) = x^5 + 10x^3 + 3x^2 + 12$. (08 Marks)
- c. Obtain triplet representation for the given sparse matrix. Write fast transpose algorithm to obtain transpose of sparse matrix.

$$\begin{bmatrix} 15 & 0 & 0 & 22 & 0 & -15 \\ 0 & 11 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 91 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 28 & 0 & 0 & 0 \end{bmatrix}$$

(08 Marks)

Module-2

- 3 a. How recursion uses stack during its execution. Give algorithm to simulate Tower of Hanoi. Trace the algorithm for a total of 3 disc which are placed in source pole. (06 Marks)
- b. Write C routines to implement operations on stack. Also incorporate useful routines to check the stack status for full and empty. Also include global declarations. (07 Marks)
- c. Write algorithm to convert infix expression to prefix form. Apply the algorithm to obtain equivalent prefix form. Infix expression : $6 * 2 \wedge 2 \wedge 3 / (9 - 3)$ (07 Marks)

OR

- 4 a. Design circular queue using dynamically allocated arrays. Give steps to relocate elements in dynamic array for proper insertion and deletion. (04 Marks)
- b. With the help of algorithm, evaluate the postfix expression $6223 \wedge \wedge * 3 /$ using stack. (08 Marks)
- c. What is the advantage circular queue over ordinary queue? Give ADT to perform various operations on circular queue. Also give ADTs to check for empty and full. (08 Marks)

Module-3

- 5 a. Give structure representation in C to create a single linked list. Give C routine to implement following operations on SLL:
- Create SLL of integer data
 - Insert a node at rear end
 - Delete a node from front end
 - Display all nodes neatly
 - Search for a suitable data in SLL and display appropriate message. (12 Marks)
- b. What is the advantage of doubly linked list? Give suitable steps to insert a node between A and B (consider A is NULL, B is NULL and A & B are not NULL) in SLL. (08 Marks)

OR

- 6 a. Write the node representation of the linked representation of a polynomial. Also give algorithm to perform addition on two polynomials. (10 Marks)
- b. Differentiate between SLL, DLL, circular linked list and header linked list. Give algorithm to insert a node in circular linked list and traverse the list. (10 Marks)

Module-4

- 7 a. Define tree. For the given tree, explain terminologies and write the answer:
- Degree
 - Non terminal
 - Sibling
 - Ancestor
 - Level
 - Height

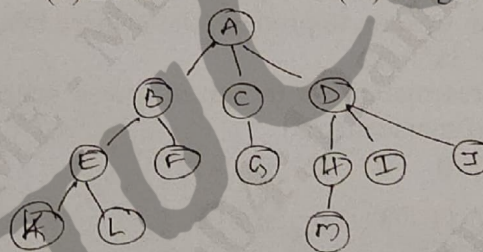


Fig.Q7(a)

(06 Marks)

- b. Give C routine to create BST for the data 12, 0, -90, 5, 3, 10, 0, 8, 18. Give 3 traversals of BST constructed from above data. (07 Marks)
- c. Given in order sequence DJHBEAFICG and post order sequence JHDEBIFGCA, construct binary tree and give pre-order traversal. (07 Marks)

OR

- 8 a. Give array and linked list representation for the binary tree.

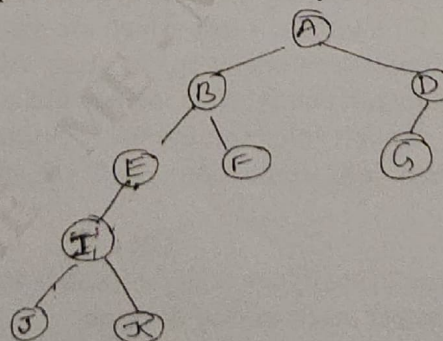


Fig.Q8

(06 Marks)

- b. Write iterative and recursive search function to search a key in BST. (08 Marks)
- c. Draw a binary tree for the following expression $3 + 4 * (7 - 6) / 4 + 3$. Traverse the tree and obtain pre-order and post order expression. (06 Marks)

Module-5

- 9 a. For the given graph show adjacency matrix and adjacency list representation.

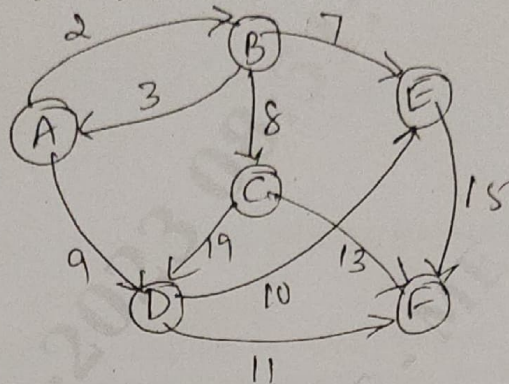


Fig.Q9(a)

- b. Write BSF and DFS algorithm for graph traversal.
 c. Write a note on AVL tree.

(06 Marks)

(10 Marks)

(04 Marks)

OR

- 10 a. What is hashing? Explain different hashing function with suitable numerical example.

(08 Marks)

- b. What is collision? Explain the method to resolve collision with suitable algorithm of linear probing. Insert keys 72, 27, 36, 24, 63, 81, 92, 101 into table [size = 10].

(08 Marks)

(04 Marks)

- c. Write a note on B-tree.

